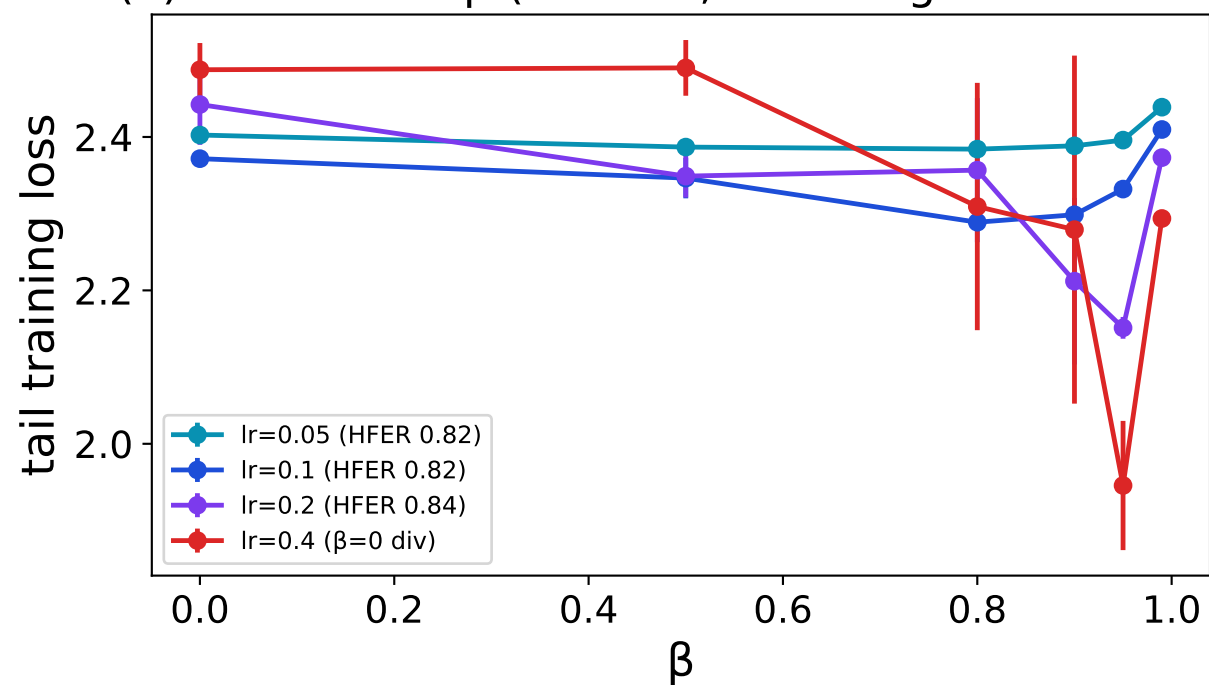
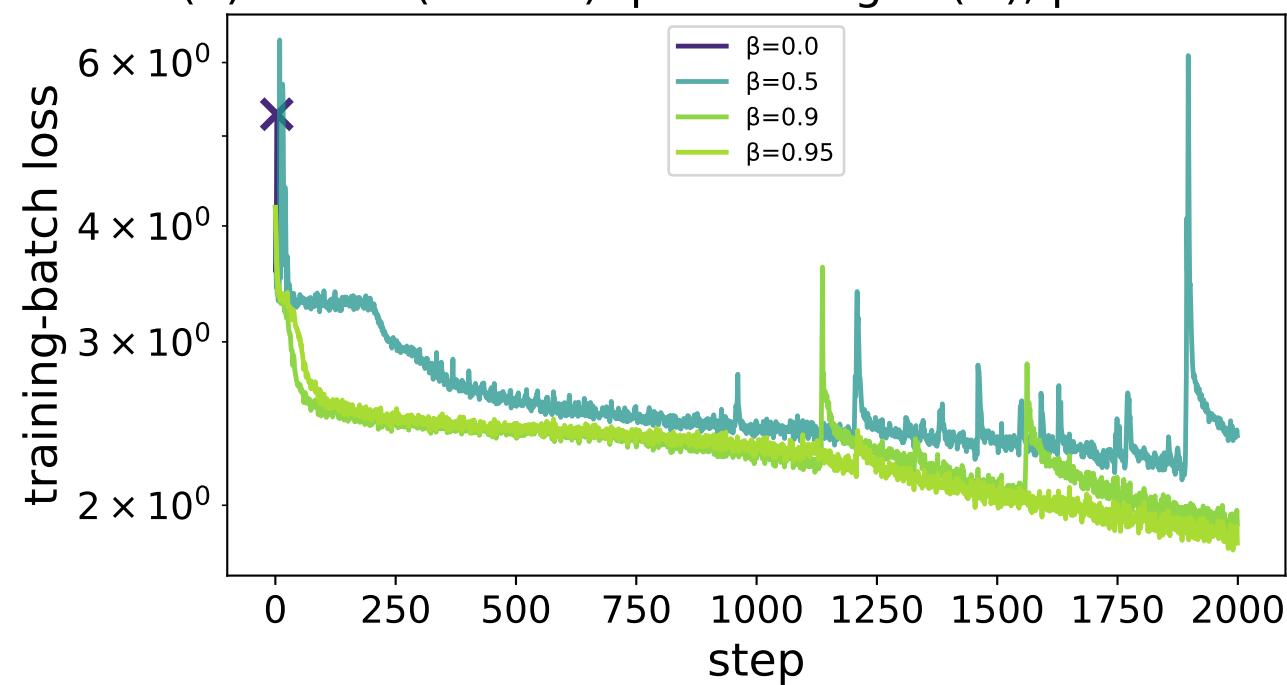
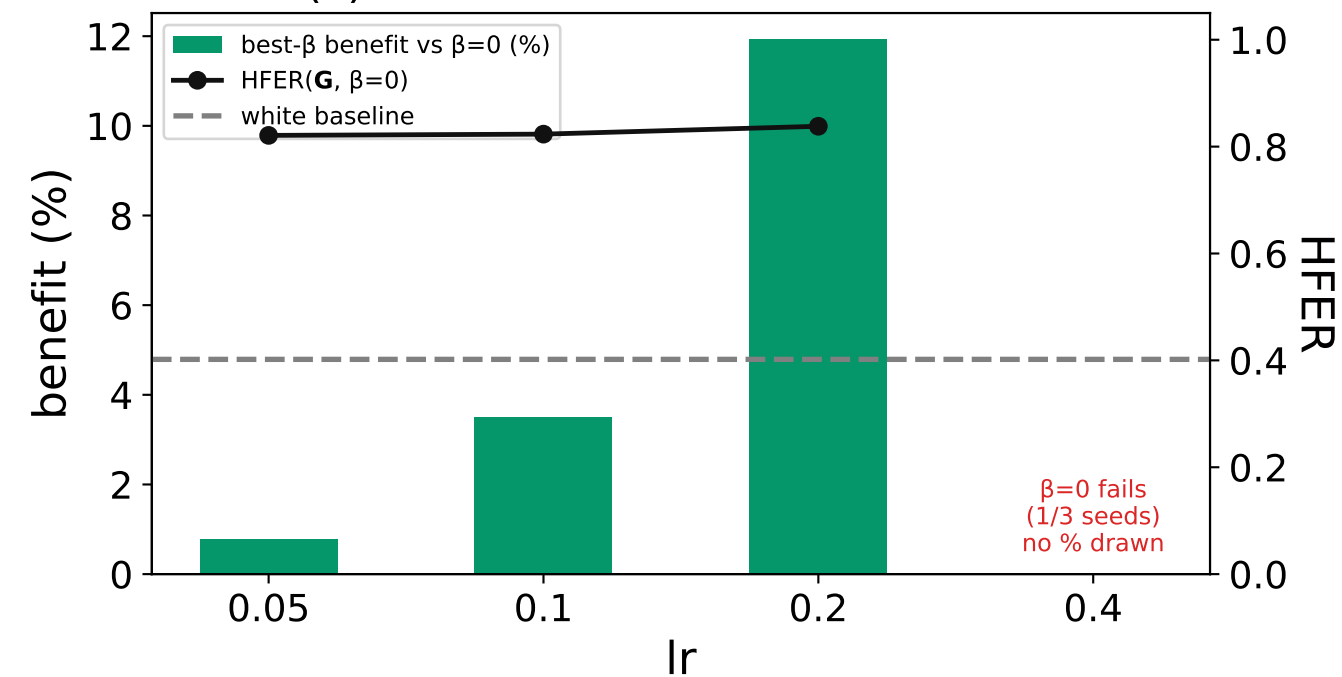


(a) SGDM sweep (3 seeds; surviving-seed means)

(b) lr=0.4 (seed 0): $\beta=0$ diverges ($\times$), $\beta \geq 0.9$ trains

(c) benefit and stream HFER vs lr



(d) Muon closed loop at scale

